Started on	Monday, 28 September 2026, 9:17 PM
State	Finished
Completed on	Monday, 28 September 2026, 9:40 PM
Time taken	23 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void merge(int a[], int left, int mid, int right) {
    int i = left, j = mid + 1, k = 0;
    int temp[right - left + 1];

    while (i <= mid && j <= right) {
        if (a[i] <= a[j])
            temp[k++] = a[i++];
        else
            temp[k++] = a[j++];
    }

    while (i <= mid)
        temp[k++] = a[i++];

    while (j <= right)
        temp[k++] = a[j++];

    for (i = left, k = 0; i <= right; i++, k++)
        a[i] = temp[k];
}

void mergeSort(int a[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;

        mergeSort(a, left, mid);
        mergeSort(a, mid + 1, right);
        merge(a, left, mid, right);
    }
}

int main() {
    int n;
    scanf("%d", &n);

    int a[n], b[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (int i = 0; i < n; i++)
        scanf("%d", &b[i]);

    mergeSort(a, 0, n - 1);
    mergeSort(b, 0, n - 1);

    long long sum = 0;

    for (int i = 0; i < n; i++) {
        sum += (long long)a[i] * b[n - 1 - i];
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.